THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The number of seconds in 7 days is: 1

(a) 6,05,800 **(b)** 60,480 **(c)** 6,04,800 **(d)** 6,01,200

2. The value of $(125 - 5 \times (9 + 11)) \div 5 \times 3$ is: 1

(a) 13 **(b)** 17 **(c)** 15 **(d)** 25

3. $0.2 \times 0.3 \times 0.4 =$ 1

(a) 2.4 **(b)** 0.24 **(c)** 0.0024 **(d)** 0.024

4. If $a = 3$ and $b = 4$, the value of $5a^2 - 6b$ is: 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 20

5. The angle between the bisectors of the two angles of a linear pair is: 1

(a) 90° **(b)** 60° **(c)** 45° **(d)** 180°

6. The smallest 3-digit number divisible by 9 is: 1

(a) 109 **(b)** 107 **(c)** 108 **(d)** 117

7. The number of triangles with whole-number sides (in cm) and perimeter 9 cm is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

8. $\frac{3}{4} \div (\frac{3}{2} \div \frac{1}{2})$ equals: 1

(a) 1 **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{8}$ **(d)** $\frac{1}{4}$

9. The number of zeros in 25,000 × 400 is: 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

10. The value of 'twice the sum of 7 and 5, decreased by 9' is: 1

(a) 6 **(b)** 24 **(c)** 33 **(d)** 15

11. How many times does 0.05 go into 2.4? 1

(a) 47 **(b)** 49 **(c)** 48 **(d)** 50

12. Simplify $5(x + 2) - 3(x - 1)$: 1

(a) $2x + 7$ **(b)** $2x - 7$ **(c)** $8x + 13$ **(d)** $2x + 13$

13. Two supplementary angles are in the ratio $1 : 2$. The larger angle is: 1

(a) 130° **(b)** 120° **(c)** 140° **(d)** 110°

14. The largest 4-digit number with all digits different that is divisible by 9 is: 1

(a) 9874 **(b)** 9872 **(c)** 9875 **(d)** 9873

15. The exterior angle at the base of an isosceles triangle is 110°. The apex angle is: 1

(a) 60° **(b)** 30° **(c)** 50° **(d)** 40°

16. $\frac{2}{5}$ of a number is 18. Three times the number is: 1

(a) 270 **(b)** 135 **(c)** $135\frac{1}{2}$ **(d)** $67\frac{1}{2}$

17. 1 crore = 1,00,00,000; the number of crores in 25,00,00,000 is: 1

(a) 2500 **(b)** 2.5 **(c)** 250 **(d)** 25

18. Compare 14 × 9 and 13 × 10. The first minus the second is: 1

(a) -2 **(b)** -3 **(c)** -4 **(d)** -5

19. 0.6 of a number is 4.2. The number is: 1

(a) 9 **(b)** 8 **(c)** 7 **(d)** 6

20. The perimeter of a rectangle with sides $5x + 4$ and $3x - 1$ is: 1

(a) $16x + 6$ **(b)** $30x$ **(c)** $8x + 3$ **(d)** $4x + 10$

21. The supplement of the complement of 20° is: 1

(a) 100° **(b)** 120° **(c)** 110° **(d)** 130°

22. Kaprekar's routine on 2111 reaches 6174 after how many steps? 1

(a) 7 **(b)** 5 **(c)** 6 **(d)** 4

23. In $\triangle ABC$, $\angle A = \angle B + \angle C$. Then $\angle A$ is: 1

(a) 45° **(b)** 120° **(c)** 90° **(d)** 60°

24. A pipe fills $\frac{2}{5}$ of a tank in 12 minutes. The time to fill the whole tank (minutes) is: 1

(a) 60 **(b)** 30 **(c)** 15 **(d)** $30\frac{1}{2}$

25. A tank holds 1,25,000 litres. How many such tanks hold 60,00,000 litres? 1

(a) 49 **(b)** 48 **(c)** 58 **(d)** 47

26. The value of 23 × 99 can be found as 23 × 100 - 23 = : 1

(a) 2254 **(b)** 2300 **(c)** 2277 **(d)** 2377

27. $(2.5 + 1.75) \times 0.4 =$ 1

(a) 1.6 **(b)** 1.7 **(c)** 2.7 **(d)** 1.8

28. If $n = 10$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 56 **(b)** 55 **(c)** 54 **(d)** 57

29. The reflex angle that goes with an angle of 70° (so that together they make a complete turn) is: 1

(a) 280° **(b)** 290° **(c)** 310° **(d)** 300°

30. The Collatz sequence from 12 needs how many steps to reach 1? 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

31. The angles of a triangle are $(2x)^\circ$, $(3x + 5)^\circ$ and $(4x - 50)^\circ$. The largest angle is: 1

(a) 70° **(b)** 80° **(c)** 90° **(d)** 100°

32. $\frac{1}{3}$ of a wall is already painted. Then $\frac{1}{2}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{7}{12}$

33. The smallest number that must be added to 34,56,789 to get the next lakh is: 1

(a) 43,211 **(b)** 44,211 **(c)** 43,111 **(d)** 99,999

34. 12 × 38 + 12 × 62 is equal to 12 × ?. The missing number is: 1

(a) 112 **(b)** 100 **(c)** 99 **(d)** 101

35. Which of these is closest to 1? 1

(a) 1.2 **(b)** 1.08 **(c)** 0.95 **(d)** 0.89

36. Which expression is always even for a whole number n? 1

(a) $2n + 4$ **(b)** $n + 2$ **(c)** $2n + 1$ **(d)** $3n$

37. Two complementary angles differ by $20°$. The larger angle is: 1

(a) $45°$ **(b)** $65°$ **(c)** $55°$ **(d)** $75°$

38. For every whole number n, $n(n + 1)$ is: 1

(a) even **(b)** a multiple of 3 **(c)** odd **(d)** a perfect square

39. Sides 11 cm, 6 cm and 5 cm are given. Which statement is true? 1

(a) only a right triangle is possible **(b)** only an equilateral triangle is possible
(c) no triangle can be formed **(d)** a triangle can be formed

40. If $0 < x < 1$, then $\frac{1}{x}$ is: 1

(a) negative **(b)** greater than 1 **(c)** less than 1 **(d)** equal to 1

41. **Assertion (A):** 1 crore = 100 lakh. 1
Reason (R): 1 lakh = 1,00,000 and 1 crore = 1,00,00,000.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** $5 + 3 \times 4 = 17$. 1
Reason (R): In an expression without brackets, multiplication is done before addition.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** 0.5 is greater than 0.45. 1
Reason (R): Comparing digit by digit from the left, 5 tenths is more than 4 tenths.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** 'Five more than three times n' is written as 3n + 5. 1
Reason (R): Three times n is written as 3n.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The complement of 35° is 55°. 1
Reason (R): Two angles are complementary if their sum is 90°.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The sum of two odd numbers is even. 1
Reason (R): An odd number is of the form 2n + 1, so the sum of two odd numbers is 2(m + n + 1).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** A triangle cannot be formed with sides 3 cm, 4 cm and 8 cm. 1
Reason (R): The sum of any two sides of a triangle must be greater than the third side.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$. 1
Reason (R): The product of two fractions is (product of numerators) / (product of denominators).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** 1 million = 10 lakh. 1
Reason (R): 1 billion = 1000 million.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** $(3 + 4) \times 2 = 14$. 1
Reason (R): $3 + 4 \times 2 = 11$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**